

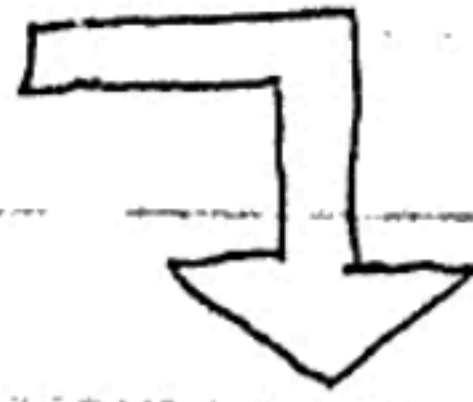
Exercise 01 - SQL Fundamentals.

1) Retrieve all columns from the employees table

SELECT *

FROM employees;

employees table.



id	first-name	last-name	department	Salary	hire-date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2021-09-30	Johannesburg

2) Find all unique departments

SELECT DISTINCT departments

FROM employees;



department
IT
HR
Finance
Marketing
IT

3) Retrieve first and last names ordered by salary descending.

SELECT first-name, last-name

FROM employees

ORDER BY salary DESC;



first-name	last-name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. Retrieve the top 3 highest-paid employees.

```
SELECT id, first-name, last-name, salary  
FROM employees  
ORDER BY salary DESC  
LIMIT 3;
```

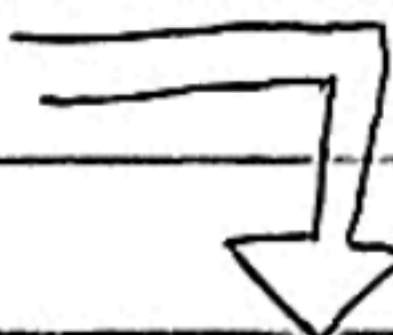


id	first-name	last-name	salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

02 Filtering with WHERE

5 Find employees in the IT department.

```
SELECT *  
FROM employees  
WHERE department = 'IT';
```



id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6 Find employees in Finance with salary > 60000.

```
SELECT *  
FROM employees  
WHERE department = 'Finance' AND  
salary > 60000;
```



id	first-name	last-name	department	salary	hire-date	city
3	Cathy	Zulu	Finance	65000	2018-01-18	Durban.

7. Find employees in HR or Marketing.
 SELECT id, first_name, last_name, department
 FROM employees
 WHERE department IN ('HR', 'Marketing');



id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. Find employees not in IT
 SELECT id, first_name, last_name, department
 FROM employees
 WHERE NOT department = 'IT';



id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. Find employees in IT, HR, or Finance using IN
 SELECT id, first_name, last_name, department
 FROM employees
 WHERE department IN ('IT', 'HR', 'Finance');



id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance

10. Find employees in IT with salary > 65000 and city Johannesburg.

SELECT id, first_name, last_name, department, salary, city

FROM employees

WHERE department = 'IT'

AND salary > 65000

AND city = 'Johannesburg';



id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg